

LE DINH HOANG VU

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SUMMARY

Computer Science student specializing in Computer Vision with hands-on experience deploying edge vision prototypes on ESP32-CAM using C/C++ and TensorFlow Lite Micro. Built AIoT workflows combining offline inference, device alerts, and network-resilient logging.

PROJECT & RESEARCH EXPERIENCE

AIoT Drowsiness Detection System - Edge AI for Driver Safety ([Repository](#)) Ho Chi Minh City, Vietnam
Academic prototype for driver warning without continuous Internet dependency. Oct 2025 - Dec 2025

- Built an ESP32-CAM/OV2640 pipeline with ESP-DL face detection, dynamic eye-region cropping, and 96x96 image preprocessing.
- Deployed an INT8 FOMO MobileNetV2 0.35x model with TensorFlow Lite Micro and PSRAM buffers for offline eye-state inference.
- Implemented sustained-eye-closure filtering with UART audio warnings and HTTPS Telegram photo/text alerts.
- Integrated a FreeRTOS upload task with offline log buffering, Wi-Fi recovery, and NTP timestamps for intermittent connectivity.

CropVision AI - AI Software Platform for Agriculture ([Repository](#)) Ho Chi Minh City, Vietnam
Graduation Project, In Progress - software services supporting crop disease inference. May 2026 - Present

- Developed a FastAPI YOLOv8 service with image validation, bounding boxes, and confidence outputs for REST-based inference.
- Integrated Express.js, JWT, PostgreSQL scan history, and Docker Compose to connect uploads with persisted AI results.

EDUCATION

University of Science, VNU-HCM Ho Chi Minh City, Vietnam
Bachelor of Science in Computer Science, Specialization in Computer Vision Sep 2022 - Jun 2026
(Expected)

GPA: 3.1/4.0

TOEIC: 650 LR – 290 SW

PUBLICATIONS

Nguyen Trung Kien, **Le Dinh Hoang Vu**, Vo Hoai Viet. "Detection of Common Crop Leaf Diseases in Vietnam Using an Explainable YOLO Architecture." *Vietnam Journal of Science and Technology, Series B*. Published Jan 2026. [DOI: 10.31276/VJST.2025.3727](#)

ADDITIONAL INFORMATION

Languages: C/C++, Python, JavaScript/TypeScript

Technologies: ESP32-CAM, Jupiter, Edge Impulse/FOMO, Arduino OpenCV, FastAPI, Docker, Git

Language Skill: Reading English Document influent